

Self-organized porous titanium oxide prepared in $\text{Na}_2\text{SO}_4/\text{NaF}$ electrolytes

Jan M. Macak, K. Sirotna¹, P. Schmuki^{*,2}

University of Erlangen-Nuremberg, Department of Materials Science, Chair for Surface Science and Corrosion, D-91058 Erlangen, Germany

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Abstract

The anodic formation of nanoporous TiO_2 on titanium was investigated in Na_2SO_4 electrolytes containing low concentrations of NaF (0.1–1 wt.%). It was found that under optimized electrolyte conditions and extended polarization, a self-organized nanostructure consisting of porous TiO_2 is obtained. The porous structure is arranged in sections of arrays with single pore diameters of typically 100 nm and an average spacing of 150 nm. The pores are open at the top and covered by oxide at the bottom. Compared with earlier work, we show that using a neutral NaF electrolyte significantly thicker porous layers can be obtained than in acidic solutions.

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1. Introduction

In recent years the electrochemical formation of self-organized porous structures has been reported for materials such as aluminium [1–3], silicon [4,5] or InP [6]. Porous structures can be electrochemically formed under very defined anodizing conditions. Namely a specific voltage, electrolyte composition and temperature. For titanium, anodization in various electrolytes has been studied for many decades. However, self-organized porosification has recently been found only in a number of HF-containing electrolytes such as aqueous HF electrolytes [7,8], chromic acid–HF mixtures [9] or H_2SO_4 –HF mixtures [10]. These self-organized porous TiO_2 structures bear a large potential in view of several applications. This particularly because TiO_2 possesses a range of specific functional properties as for example bio-

[11] and gas- [12] sensing, self-cleaning [13] or photoelectrochemical conversion [14]. Furthermore the nanostructures may be used as a template (for the deposition of other materials into the pores, followed by selective etching of the TiO_2 mold) in analogy to aluminium [15,16] and may find application for the production of magnetic recording media [17,18], functional electrodes [19,20], photonic crystals [21] and for waveguides [22]. An advantage of this processing approach is that self assembly can be regarded as a smart nano approach, i.e. it does not require active manipulation on the nanoscale such as for example when STM or AFM based techniques are used.

Previous work [10] showed that the porosification process in HF is based on a steady state situation that is the pore formation reaction at the inner (metal) interface and a chemical dissolution of the oxide at the outer interface. As a result only a limiting thickness of the porous layer can be achieved. Common to the previous works is that this limiting layer thickness is in the range of approximately 500–800 nm [7–10].

However, in parallel with the present work it was reported [23] that the high aspect ratio TiO_2 nanotubes can be formed in $\text{KF}/\text{SO}_4^{2-}$ electrolytes containing citric acid. More recently we showed that fully self-organized high aspect

* Corresponding author. Tel.: +49 9131 852 7575; fax: +49 9131 852 7582.

E-mail address: schmuki@ww.uni-erlangen.de (P. Schmuki).

¹ Permanent address: Institut of Chemical Technology Prague, Department of Power Engineering, CZ-16628 Prague, Czech Republic.

² ISE member.

ratio TiO_2 nanotubes can be obtained for the first time in $\text{NH}_4\text{F}/(\text{NH}_4)_2\text{SO}_4$ electrolytes, if the electrochemical conditions are adequately tuned [24].

In the present work we explore the possibility to achieve self-organized porous TiO_2 structures in neutral $\text{NaF}/\text{Na}_2\text{SO}_4$ mixtures. This with the aim to achieve a comparison and expansion of earlier work in view of feasibility and effects of the electrolyte on the pore structure.

2. Experimental

Titanium samples (99.6% purity, Goodfellow, England), foils of 0.1 mm of thickness, were degreased by sonicating in acetone, isopropanol and methanol, followed by rinsing with deionized (DI) water and drying in a nitrogen stream without any additional surface polishing.

For electrochemical experiments, the samples were contacted and then pressed against an O-ring in a electrochemical cell, leaving 1 cm^2 exposed to the electrolyte. The electrochemical setup consisted of a conventional three-electrode configuration with a platinum gauze as a counter electrode and a Haber–Luggin capillary with Ag/AgCl (1M KCl) electrode as a reference electrode. Electrochemical experiments were carried out using high-voltage potentiostat Jaissle IMP 88 PC–200 V, connected to digital multimeter (Keithley 2000) interfaced to a computer. All anodization experiments were carried out at room temperature.

As an electrolyte for electrochemical experiments we used 1M Na_2SO_4 solutions with the addition of different amounts of NaF (0.1–1 wt.%). All solutions were prepared from reagent grade chemicals and DI water. If not denoted otherwise, the electrochemical treatment consisted of a potential ramp from the open-circuit potential (OCP) to 20 V with a sweep rate of 0.1 V/s followed by holding the samples at 20 V for different times. After the formation of the porous structures, the samples were immediately and intensively rinsed with DI water and then dried with a nitrogen stream.

For the structural and morphological characterization of the anodized samples, top view, cross-sectional and bottom view electron microscope (SEM) observations and direct SEM cross-sectional thickness measurements were carried out using a Hitachi FE-SEM S4800 or Leo 1530 SEM. The cross-section images were taken from mechanically bent samples, where a lift-off of the porous layer occurred. These samples could be directly used for SEM layer thickness measurements.

Dissolution rate experiments with anodically formed TiO_2 layers were performed using samples (thickness 2 mm, 99.6%, Goodfellow, England) that were treated with 1200 SiC grinding paper and then polished with $1.4\text{ }\mu\text{m}$ diamond paste (both Struers, Denmark). These samples were anodized in H_2SO_4 (1 M) at 20 V for 5 min. This led to the formation of a compact 50 nm thick anodic oxide layer (using a growth factor of 2.5 nm/V [25]). The samples then were etched for vari-

ous times in a solution containing 1 M Na_2SO_4 and 0.5 wt.% NaF adjusted to different pH-values 2, 4 and 5.3 using adequate additions of 1 M H_2SO_4 to determine the dissolution rate as a function of pH.

The composition and thickness of the samples were characterized by X-ray photoelectron spectroscopy and by Auger electron spectroscopy. The thickness of the TiO_2 layers was detected from depth sputter profiles (sputter times) determined by X-ray photoelectron spectroscopy (PHI 5600 XPS) acquired by Ar^+ sputtering at 3.2 kV using the Ti peak at 459 eV, the O peak at 528 eV and the F peak at 685 eV. The sputter rate was obtained from calibration with SiO_2 . XPS depth profiles were acquired after 30 and 60 s after immersing to the solution. Auger electron spectroscopy (AES) characterization was carried out using a PHI 670 AES (utilizing the Ti peak at 385 eV and the O peak at 508 eV).

3. Results and discussion

Within this work electrochemical anodization was performed by holding a potential of 20 V constant after a potential ramp from the open-circuit potential. Preliminary experiments showed that best results in view of self-organization of the pores, such as shown in Figs. 1 and 2, are obtained, within the frame of the conditions used here, at the potential of 20 V (lower potentials lead to less ordered structure formation, significantly higher potentials lead to breakdown [26] and sparking [27]).

In order to elucidate the influence of some other key parameters on the pore morphology, a set of experiments was performed with different NaF concentrations and polarization times. SEM characterization was performed after the experiments. Anodization in Na_2SO_4 solution containing 0.1 and 0.3 wt.% of NaF results in a relatively heterogenous etch structure, which is not uniform and far from being organized. The pore formation in this case does not take place on the entire surface, there are islands that are preferentially etched, these etch patterns cover approximately half of the surface area. Additionally there are cloudy precipitates on the rest of the exposed area (additional XPS measurements on the precipitated layer composition showed that the insoluble layer consist of Ti-, O- and F-species). Anodization in Na_2SO_4 (1 M) solution containing NaF in the range of 0.5–1 wt.% results in far more homogenous porous structures. Electrolytes with NaF concentration higher than 1 wt.% cannot be produced due to the limited solubility of NaF in aqueous solution of 1 M Na_2SO_4 . Fig. 1 shows the surface morphology of samples anodized in electrolytes containing 0.5 wt.% NaF for various times. The porous surface shows a uniform, regular structure without any significant remains of precipitates after a certain time. The sample oxidized for 10 min (Fig. 1a) shows typically a very rough surface with apparent patches of non-porous structure or a precipitate cover. The sample oxidized for 1 h (Fig. 1b) shows a large area covered with

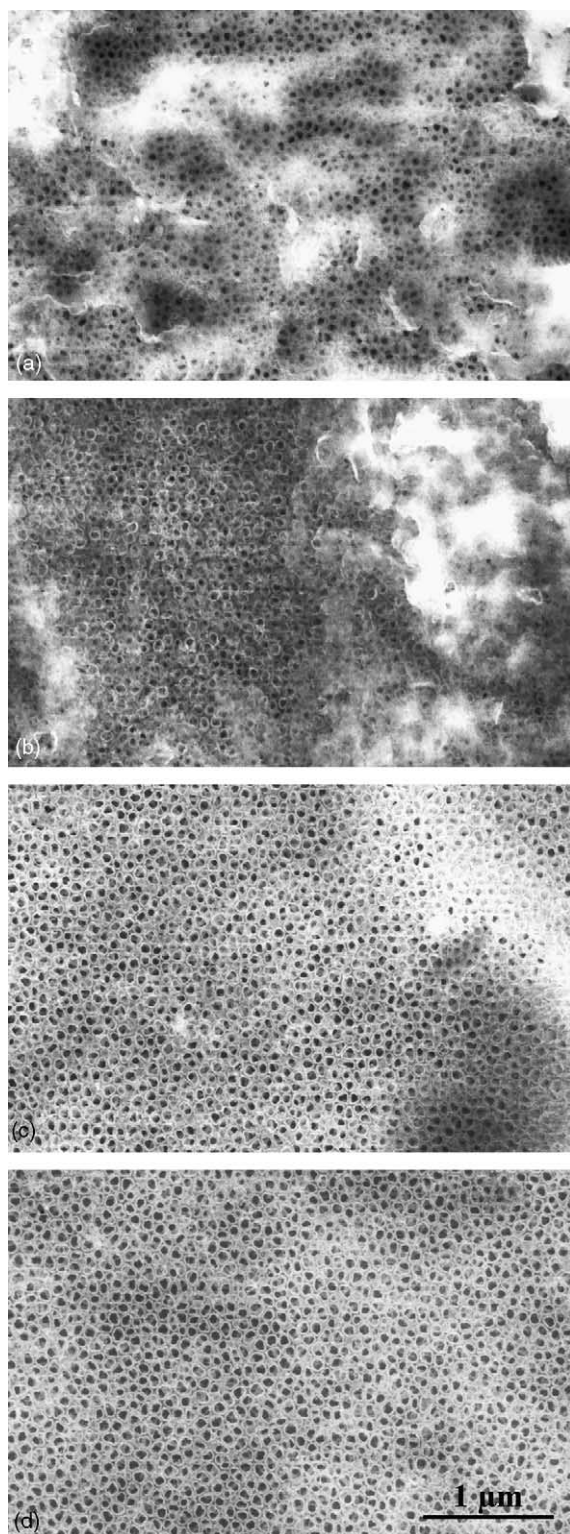


Fig. 1. SEM images of titanium anodized at 20 V in Na_2SO_4 (1 M) + NaF 0.5 wt.% for (a) 10 min, (b) 1 h, (c) 2 h and (d) 6 h.

the porous structure. On a large scale fewer non-porositized islands and cloudy precipitates were found than in the previous case (Fig. 1a). After 2 h (Fig. 1c) well structured areas of pores cover almost the entire surface. The amount of precip-

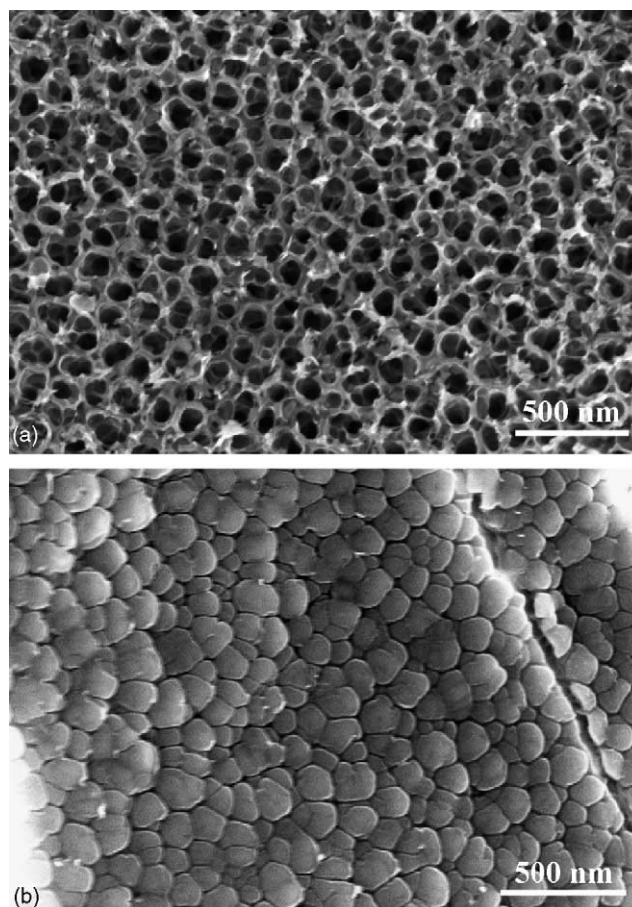


Fig. 2. SEM top- (a) and the bottom-view (b) of the porous structure on sample anodized for 6 h in Na_2SO_4 (1 M) + 0.5 wt.% NaF, average pore diameter is 100 nm. The SEM bottom-view was taken from mechanically cracked sample, where pieces of the porous layer were laying upside down on the surface.

itates on the top of the porous structure is negligible. After 6 h of anodization (Fig. 1d), a regular porous structure has formed consisting of organized pores with single pore diameters of approximately 100 nm and with an average spacing of 150 nm. The sequence in Fig. 1 illustrates that for a self-organization considerable time has to pass, i.e. several hours of pore drilling at the inner interface combined with oxide dissolution at the outer interface must have passed.

Fig. 2 shows again SEM images of the self-organized porous titanium oxide layer of Fig. 1 after 6 h and compares the top (Fig. 2a) and the bottom (Fig. 2b) morphology of the porous structure. The bottom morphology (Fig. 2b) was taken from a mechanically cracked sample. It is clear from this figure that porous structure is open on the top of the layer while on the bottom of the structure are the tubes closed. It is noteworthy to point out some differences to the well investigated case of Al_2O_3 pores. A top open and bottom closed morphology is present also in the Al case however for Al thick pore walls are typically observed [28] while in the present case the pore walls are only in the order of 10–20 nm thick. By comparing Fig. 2a and b it can be seen that the pore diameter

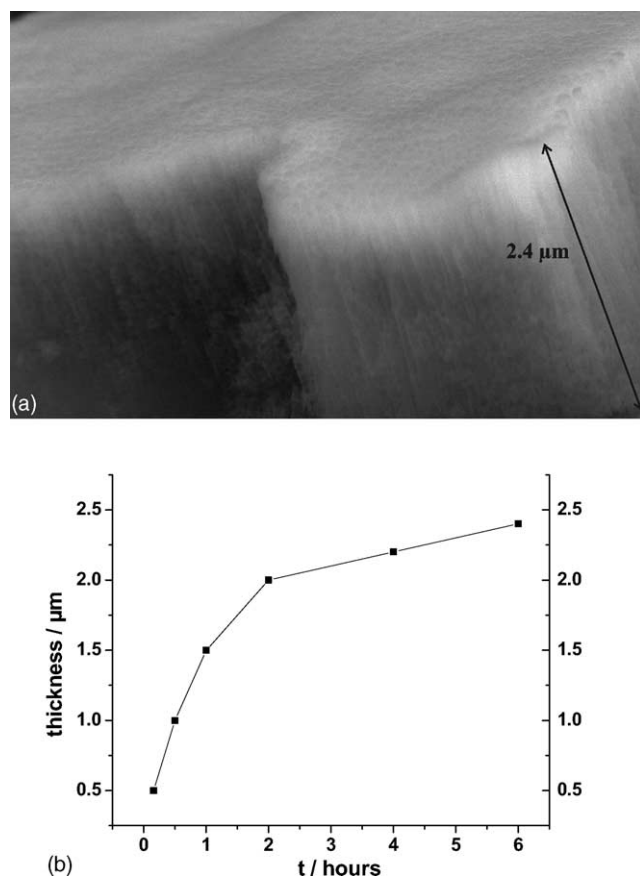


Fig. 3. (a) FE-SEM cross-sectional view of a piece of the porous structure which cracked off after folding a sample anodized for 6 h in Na_2SO_4 (1 M) + 0.5 wt.% NaF. (b) The thickness of the porous layer as a function of the anodization time (10 and 30 min; 1, 2, 4 and 6 h) for polarization in Na_2SO_4 (1 M) + 0.5 wt.% NaF.

of approximately 100 nm is reflected also in the bottom morphology where hillock spacing also of about 100 nm can be observed. This suggests that the lateral spacing of the ordering phenomenon is kept constant during etching.

The top view shows that there is still some irregularity in the pore formation process. In fact some of the pores seem not to grow absolutely straight and some etch traces on the surface are apparent. This indicates that within the present work still not absolutely optimized parameters have been found.

In order to gain information on thickness of the porous layer, SEM cross-sectional views were taken from the mechanically bent samples. Fig. 3a shows a piece of the porous titanium oxide that partially cracked off after folding a sample. From the cross-sectional view it is apparent that the layer thickness is approximately 2.4 μm. Most of the structure consists of tube like sections. The same cross-sectioning procedure was applied to a series of samples that were polarized in Na_2SO_4 (1 M) + 0.5 wt.% NaF for different times. Fig. 3b shows the thickness of the porous layer as a function of the anodization time for this electrolyte. The thickness first increases steeply from about 1 μm at 30 min to 2 μm at 2 h. This is followed by a steady state with a plateau of thick-

ness values at approximately 2.4 μm reached after 6 h. Additionally it can be concluded that indeed steady state has been reached after approximately 6 h from the fact that in the current–time curves a constant current was reached. In accord with previous work [10], XPS and AES analysis revealed that the porous layer consisted in every case of approximately an atomic concentration ratio of 1:2 for Ti:O and from XPS a peak position corresponding to TiO_2 was found, this shows that the porous layer entirely consists of TiO_2 . Some traces of F were detected throughout the film, traces of N were present, but negligible.

SEM pictures taken from samples anodized in electrolytes with different concentration (0.5; 0.7; 1 wt.% NaF) show that pore size is not significantly depending on the NaF concentration. Furthermore no significant influence of polarization time was found. The main effect of time is a change of the layer thickness. It has been already described [10] that in presence of HF in the electrolyte permanent dissolution of the oxide takes place at the solution interface combined in the same time with oxide formation at the metal interface. In other words, the present findings are in line with the presumption that under sufficiently slow oxide etching conditions the porous layer may grow to a substantial thickness (in our case 2.4 μm) and that a steady-state situation occurs as soon as the dissolution rate is equal to rate of oxide formation. Another very interesting finding is that during potentiostatic polarization current oscillations are observed (as previously reported for the HF case [10]). Fig. 4a shows current transients recorded while holding the potential at 20 V constant after a potential ramp (0.1 V s^{-1}) in 1 M Na_2SO_4 electrolyte with and without addition of NaF (0.5 wt.%). In NaF-free Na_2SO_4 electrolyte the current drops rapidly, typical for a classical high-field passivation behavior that leads to the formation of a compact oxide film [29]. As apparent, if NaF is added to the Na_2SO_4 electrolyte, the current transient clearly deviates from the NaF-free Na_2SO_4 curve. Such curve shapes, showing a decrease/increase/decrease sequence, have been already observed for self-organized formation of porous aluminium oxide [22] and have been assigned to different stages of a self-ordering process. Furthermore, it is evident that regular current oscillations in presence of NaF occur. These events remain active during the entire polarization time. Fig. 4b shows a detailed SEM cross-section of the sample, whose current transient (0.5 wt.% NaF) is shown in Fig. 4a. It can be clearly seen that the pore tubes have rings (ripples) on their walls. These rings on each single tube could be formed due to the current oscillations, which occur during the electrochemical experiment or oppositely these current fluctuations may be caused by the formation of these rings. That there is a correlation between rings and oscillations is supported by the fact that converting the current oscillation frequency ($\sim 1 \text{ min}^{-1}$) into a dissolution length scale results in approximately 20 nm (i.e. comparable to the 25 nm spacing observed in Fig. 4).

The oxide dissolution occurs due to the formation of a soluble hexafluorotitanium complex $[\text{TiF}_6]^{2-}$, which goes into solution. In fact, in acidic solution the main part of the current

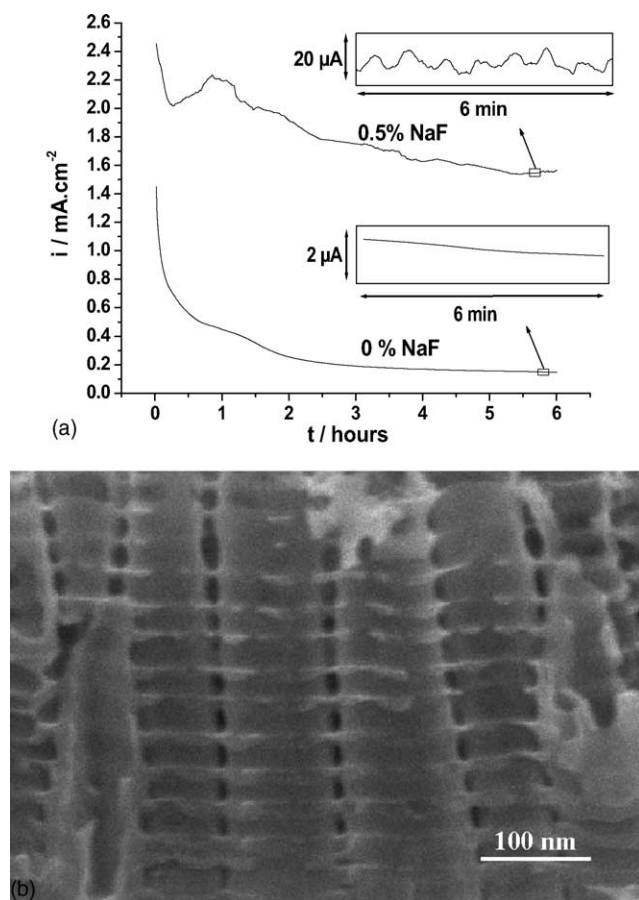


Fig. 4. (a) Current transients recorded at 20 V during anodization in 1 M Na_2SO_4 with and without 0.5 wt.% NaF addition. Taken after a potential sweep from OCP to 20 V with 0.1 V s^{-1} . (b) SEM detailed cross-section image shows pore tubes with many rings on their walls. Sample anodized in Na_2SO_4 (1 M) + 0.5 wt.% NaF for 6 h. The regularity of the rings corresponds to the periodicity of current oscillations of current–time curves.

goes into this dissolution reaction [10] and only the minor part of the current is used for TiO_2 formation. Cross-sections of samples produced in HF electrolyte showed clearly considerable material (metal) loss while in the NaF based electrolytes this effect was much less pronounced.

In order to quantify the pH effect on the dissolution of TiO_2 a set of etching experiments was performed in dependence on pH and time. XPS depth profiles were acquired afterwards to determine the remaining thickness. The initial thickness (50 nm) is taken as 100% and all other values are compared with this value. The composition of solution for dissolution is chosen to simulate anodizing conditions as much as possible. Fig. 5 gives the dissolution behaviour of the anodic titanium oxide in dependence of the pH. It is clear that the highest dissolution rate is observed in case of the solution with pH = 2 with a rate of approximately 38 nm min^{-1} , whereas at pH = 5 the dissolution rate is the lowest with a rate of less than 1.5 nm min^{-1} . In other words, clearly a strong influence of the pH on the chemical dissolution rate of TiO_2 is found. Therefore is evident that for higher pH-values the

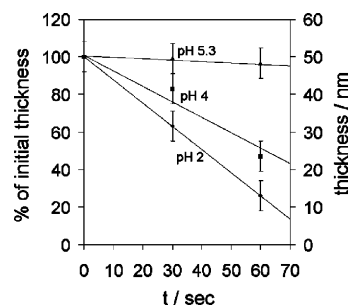


Fig. 5. Influence of pH on dissolution rate of anodic oxide in dependence of time. Samples were anodized in H_2SO_4 (1 M) at 20 V for 5 min, the thickness of formed oxide is approximately 50 nm and is taken as 100%. Samples were immersed in solution containing NaF/ Na_2SO_4 / H_2SO_4 with different pH 2, 4, 5.3 (moderated by H_2SO_4) for 30 and 60 s and thickness was compared with that initial thickness of the oxide (50 nm = 100%). Thicknesses acquired from XPS depth profiles.

chemical etching of TiO_2 is drastically slower than in acidic electrolytes.

In order to gain additional information, pH measurements were performed during electrochemical experiment. The pH-value changes during anodization because of several reasons. Due to the formation of hydrolysis products at the working electrode the pH decreases and a local acidification is established. On the counter electrode the pH increases due to hydrogen evolution and formation of OH^- species. The pH of Na_2SO_4 (1 M) + 0.5 wt.% NaF before usage is approximately 7. Experiments performed with a counter electrode sufficiently separated from the working electrode showed that at the working electrode easily pH-values of about 2 can be established. Although a conclusion of the detailed chemical speciation in dependence on pH is beyond the scope of the present work, these dissolution experiments show clearly, how the dissolution of the porous oxide layer can be affected by the solution pH-value.

4. Conclusions

We have investigated the formation of porous titanium oxide structures in Na_2SO_4 solutions containing small amounts of NaF. Under specific conditions, growth of a highly ordered porous TiO_2 layer was observed. The thickness of the oxide layer increases during long anodization time up to a limiting thickness of $2.4 \mu\text{m}$ that is reached after approximately 6 h of anodization. After this time, the initial irregular morphology has evolved to a comparably regular porous structure showing tubular features of about 100 nm diameter. The fact that a limiting thickness occurs must be associated to a competition between the formation of porous oxide and its dissolution. This dynamic equilibrium controls the time scale of the porosity development, ordering effects and the thickness of the oxide layer. The chemical dissolution rate of the oxide layers depends strongly on the pH. The observation that using NaF instead of HF leads to the growth of thicker porous layers can be associated to the fact that the chemical etching

of TiO₂ in neutral electrolytes is drastically slower than in acidic electrolytes.

While within the present work already highly regular structures were obtained we very recently showed in other work [24] that the present conditions still can be further optimized.

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